

■ Python Functions — Cheat Sheet

Today's practice: `def`, parameters, arguments, `return`, positional arguments, keyword arguments, default parameters, and basic function logic.

1. `def` — Define a Function

Use **`def`** to create a function.

```
def add(a, b):  
    return a + b
```

`add` → function name
`a, b` → parameters

2. Parameters

Parameters are the variables written inside the function definition.

```
def add(a, b):
```

`a, b` = parameters

Remember: **Parameter** = placeholder

3. Arguments

Arguments are the actual values passed when calling the function.

```
add(10, 20)
```

`10, 20` = arguments

Remember: **Argument** = actual value

4. Parameter vs Argument

Parameters appear in the function definition; arguments appear in the function call.

```
def multiply(x, y): # x, y → parameters  
    return x * y
```

```
multiply(4, 5) # 4, 5 → arguments
```

5. `return`

`return` sends a value back from the function.

```
def add(a, b):  
    return a + b
```

```
result = add(5, 3)  
print(result) # 8
```

6. `return` vs `print`

`print()` displays a value. **`return`** sends a value back so it can be stored or used.

```
def add(a, b):  
    print(a + b) # displays  
  
def add(a, b):  
    return a + b # sends value back
```

7. Positional Arguments

Arguments are matched according to their position. Position matters.

```
def student(name, age):  
    return name, age  
  
student("Loveleen", 23)  
  
name → "Loveleen"  
age → 23
```

8. Keyword Arguments

Arguments are passed using parameter names, so their order does not matter.

```
def student(name, age):  
    return name, age  
  
student(age=23, name="Loveleen")
```

9. Default Parameters

A parameter can have a default value. The default is used when no argument is provided.

```
def greet(name="Loveleen"):  
    return "Hello " + name  
  
greet() # Hello Loveleen  
greet("Aman") # Hello Aman
```

10. if/else Inside Functions

You practiced using conditions inside functions.

```
def find_max(a, b):  
    if a > b:  
        return a  
    else:  
        return b
```

11. Function + Calculation

You practiced calculations such as area and discounts.

```
def calculate_area(length, width):  
    return length * width  
  
def calculate_discount(price, discount):  
    discount_amount = price * discount / 100  
    return price - discount_amount
```

12. Function + Loop + Condition

You practiced counting vowels using a loop, condition, counter, and return.

```
def count_vowels(word):  
    count = 0  
  
    for letter in word:  
        if letter in "aeiou":  
            count += 1  
  
    return count
```

■ The Main Formula

Step	What it means
def	Creates the function
Parameter	Variable / placeholder in the function definition
Argument	Actual value passed when calling the function
return	Sends a value/result back
Result	Can store the returned value

One complete example

```
def calculate_total(price, quantity):  
    return price * quantity  
  
result = calculate_total(20, 3)  
print(result) # 60
```

Remember: def creates the function → parameters receive arguments → return sends the result back.

Today's topic: Python Functions ■